

Week 5: Geometric Foundation

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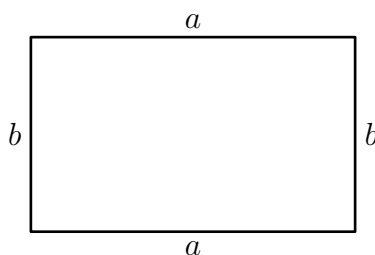
✂ Perimeter: Rectangles & Squares

Perimeter is the total distance around the outside of a shape. You can think of it as the length of a path you'd walk if you traced the edge of the shape without cutting across. If you've ever wrapped a ribbon around a box, fenced a yard, or outlined a drawing, you've used perimeter without even realizing it!

For rectangles and squares, finding the perimeter is straightforward: just add up the lengths of all the sides. But perimeter isn't always simple. Sometimes you're missing a side length and have to figure it out. Sometimes shapes are folded, cut, or made from smaller pieces, and you have to think carefully about how to compute all the missing lengths

The sum of the lengths of three sides of a rectangle is 55 centimeters. Each side is a whole number of centimeters in length. The length is 8 centimeters more than the width. Find the perimeter of the rectangle, in centimeters.

Let the side lengths of the rectangle be a and b . Consider the following diagram.



Now, we will look at the first piece of information. We know the sum of the lengths of three sides of a rectangle is 55. There are two options for this sum: either $2a + b$ or $a + 2b$.

Next, we know that a and b are whole numbers since the sides are whole number lengths. This is important, but we still don't have enough information to use this yet. Let's look at the next piece of information.

We then know that the length is 8 centimeters more than the width. In the above diagram, we can let a be the length. That is, a is larger than b . So, since the length is 8 centimeters longer than the width, then $a = 8 + b$.

Now, we can substitute this into the previous equations and solve. We have two options since either $2a + b = 55$ or $a + 2b = 55$.

- If $2a + b = 55$, then

$$\begin{aligned}
 2a + b = 55 &\iff 2(8 + b) + b = 55 \\
 &\iff 3b + 16 = 55 \\
 &\iff 3b = 39.
 \end{aligned}$$

Dividing by 3, then $b = 13$. So, $a = 21$. This seems to work. Now, let's see if the other case works.

- If $a + 2b = 55$, then

$$\begin{aligned} a + 2b = 55 &\iff (8 + b) + 2b = 55 \\ &\iff 3b = 47. \end{aligned}$$

But, if $3b = 47$, then b cannot be an integer since 3 doesn't divide 47! Therefore, this case doesn't work.

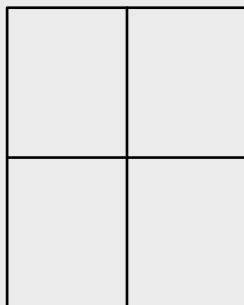
Thus, $a = 21$ and $b = 13$. Recall, we want the perimeter of the rectangle, so

$$2a + 2b = 2 \cdot 21 + 2 \cdot 13 = \boxed{68}.$$

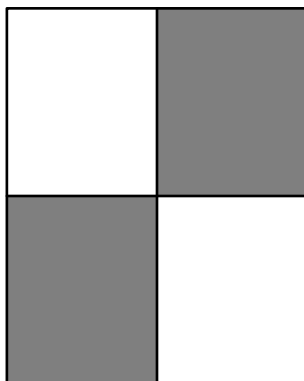
From this example, we saw how to apply a formula for finding the perimeter of a rectangle. For rectangles, the perimeter formula is simple: add the length and width, then double the result.

※ "Projecting" Lengths

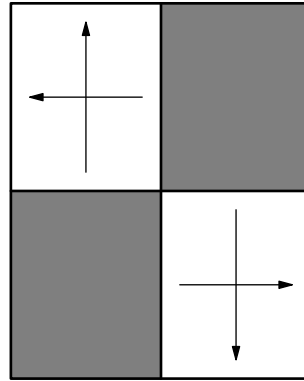
The perimeter of a rectangular piece of paper is 50 cm. It is cut into 4 congruent rectangles as shown. What is the total of the perimeters of the four smaller rectangles?



Consider two rectangles in the corner:

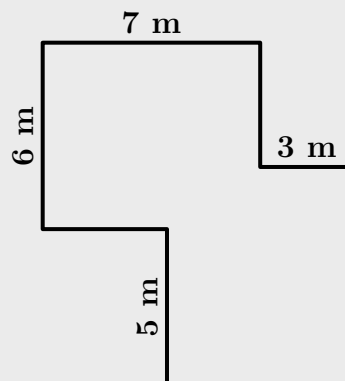


Notice the perimeter of these two rectangles is equivalent to the perimeter of the entire rectangle. This is because we can "push out" the side lengths of the two smaller rectangles to the larger rectangle. So, it would look similar to the following:

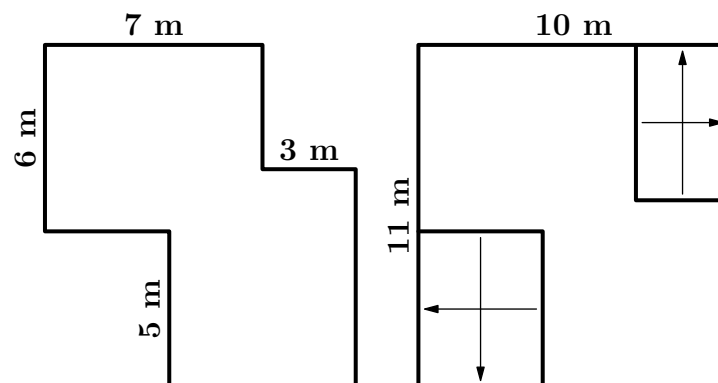


Thus, the perimeters of these two opposite rectangles sum to 50. Similarly, the perimeters of the two other rectangles in white sum to 50. Thus, the total perimeter is 100.

What is the perimeter in meters of the figure shown? All angles are right angles.



Here, we can use a trick we have learned from before by pushing the sides out:



Now, when we push the sides out, we create a 10 by 11 rectangle. The perimeter is preserved under this transformation, so we simply want to find the perimeter of the rectangle. Using the formula, then:

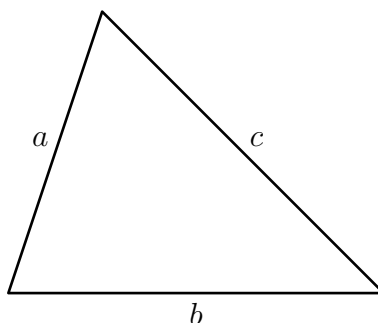
$$2 \cdot (10 + 11) = 2 \cdot 21 = \span style="border: 1px solid black; padding: 0 5px;">42.$$

※ Partial Information

In this section, we will be looking at a specific problem where you are only given partial information. In particular, you will be given a figure and you will have some information but it won't give you exactly what you want. In particular, it will give you a fraction of what you want. So, in this section, we will look at setting variables and cool manipulations to get the desired information.

Alex, Bruno, and Charles each add the lengths of two sides of the same triangle correctly. They get 27 centimeters, 35 centimeters, and 32 centimeters, respectively. Find the perimeter of the triangle, in centimeters.

Consider the following triangle and the labeling of the side lengths:



Notice, we can only obtain three possible results if we add two sides: either $a + b$, $b + c$, or $c + a$. So, we can assign the numbers 27, 35, and 32 to these sums. The order we assign them doesn't matter due to the symmetric nature of the equations. So,

$$a + b = 27, \quad b + c = 35, \quad c + a = 32.$$

Summing the equations yields $(a + b) + (b + c) + (c + a) = 27 + 35 + 32 = 94$. Now, remember we want to find $a + b + c$, the perimeter of the triangle. But, if we sum the three equations, then $(a + b) + (b + c) + (c + a) = 2(a + b + c)$. Therefore, dividing by 2 yields $a + b + c = \frac{94}{2} = \boxed{47}$.

Here, we can see the importance of setting variables and using every bit of information to come to a conclusion.

※ Area: Rectangles & Squares

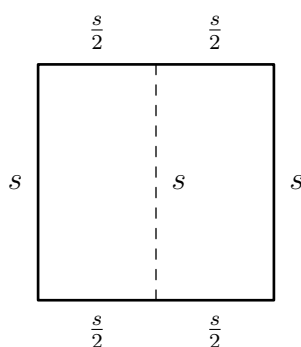
Area is one of the most fundamental concepts in geometry. But, in competitions, the problems rarely ask you to blindly plug numbers into classic formulas such as the area of a square or the area of a circle. Instead, problems have a highly creative aspect. They require you to use your imagination, apply clever concepts, and use unique techniques to simplify the problem. Many problems involve unique configurations and require creative thinking to solve. As we go through each problem, think about the different patterns that are used and how you can try to apply the techniques you learn from each problem to different ones you may have seen.

- **Area of a Square:** The area of a square with side length s is s^2 . That is, the square of the side length or the side length multiplied by the side length.

※ Setting Variables

A square piece of paper is folded in half to form a rectangle. This rectangle has a perimeter of 24 centimeters. Find the area of the original square, in square centimeters.

Generally, geometry problems are always difficult to digest without a diagram. Consider the following diagram:

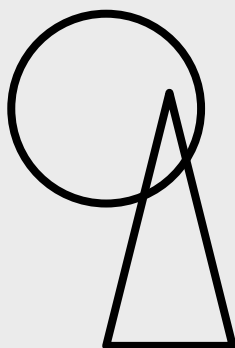


In the above configuration, the dashed line represents the folding. The two rectangles on each side of the dashed line represent the rectangles formed. Let the side length of the square be s . Then, the perimeter of the rectangles (they are identical) is the sum of its side lengths which is equivalent to:

$$\frac{s}{2} + s + \frac{s}{2} + s = 24.$$

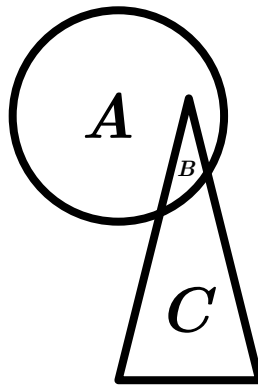
Notice, $\frac{s}{2} + s + \frac{s}{2} + s = 3s$. So, $3s = 24$. Dividing by 3 yields $s = 8$. Recall the formula for the area of the square is $s^2 = 8^2 = \boxed{64}$.

A circle and a triangle overlap as shown. The area of the circle is three times that of the triangle. If the common region is removed, then the area of the rest of the circle would be 14 square centimeters more than the area of the rest of the triangle. What is the area of the complete triangle, in square centimeters?



The key is to determine what our variables should be and how many we need. Here, Let us define three variables:

- A corresponding to the area inside the circle, but outside the triangle.
- B corresponding to the area inside both the circle and the triangle.
- C corresponding to the area inside the triangle, but outside the circle.



Now, we can use the given information from the problem to set up some equations for us to try and solve. Firstly, the area of the circle is $A + B$ while the area of the triangle is $B + C$. Since the area of the circle is three times the area of the triangle, then

$$A + B = 3(B + C).$$

Now, let's look at the next piece of information. If the common region is removed from the circle, then the area of the rest of the circle is A while the area of the rest of the triangle is C . Thus,

$$A = 14 + C.$$

It is always good to have in the back of our head what we wish to solve. Notice, we want to compute the area of the entire triangle which is $B + C$! Let's try and get rid of A in the first equation by plugging that value in. Then, we obtain:

$$\begin{aligned} A + B = 3(B + C) &\iff (14 + C) + B = 3(B + C) \\ &\iff 14 + (B + C) = 3(B + C) \\ &\iff 14 = 2(B + C) \\ &\iff 7 = B + C \end{aligned}$$

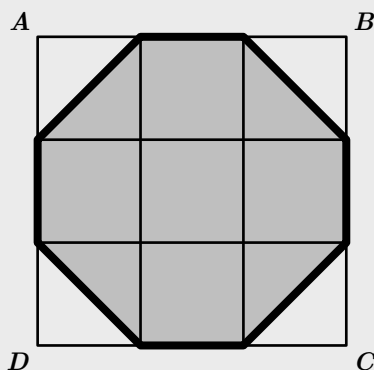
Thus, with some manipulations on what we initially had, we obtained $B + C = \boxed{7}$.

✧ Using the Grid

A grid can be a powerful visual tool to help us reason about area, shape, and structure. By carefully analyzing the grid lines and how figures align with them, we can break down complex shapes into simpler parts or compare regions more easily. This section features examples where using the grid helps us find

areas and uncover useful relationships between shapes, often without needing a formula. Keep your eyes open for symmetry, congruent shapes, and ways to count efficiently!

Square $ABCD$ is composed of nine congruent squares as shown. The area of the shaded region is 14 square centimeters. What is the area of square $ABCD$, in square centimeters?

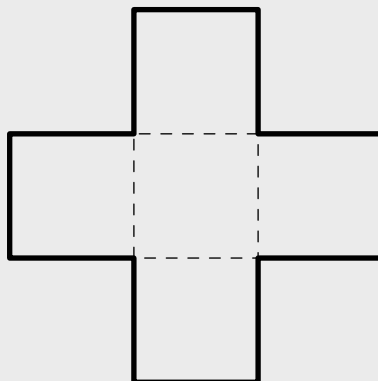


Let the area of each of the nine squares be A . Then, the total area of $ABCD$ is $9A$. Notice, the shaded region is made up of 5 whole squares, along with 4 half squares in each of the corners. Thus, the total shaded region is $5 + 4 \cdot \frac{1}{2} = 7$ squares.

Thus, 7 squares total up to an area of 14. Since a single square has area of A , then seven squares have an area of $7A$. Thus, $7A = 14$ and $A = 2$. We want to find the area of 9 total squares. To do this, we simply want to calculate the value of $9A$. Multiplying by 9 gives us $9A = 2 \cdot 9 = \boxed{18}$.

✧ Combining Area and Perimeter

This figure is made up of five congruent squares. The perimeter of the figure is 72 cm. Find the area of the figure, in square centimeters.



This figure looks very complicated at first glance. So, it is always good to try and develop a plan and dissect the problem. Notice, it is made up of five squares and we know how to find the area of a square! It is just the side length squared. So, we want to find the area of a single square, then multiply by 5. To do this, we

want to try and find a side length!

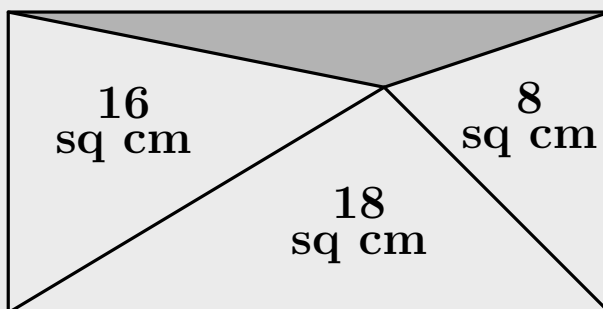
Lets now use the given information. We know the perimeter is 72 which does help give us some information on side lengths. Crucially, it is important to notice that the figure is made up of squares, so each side has the same length. Let this common length be s . Then, we can count the number of sides in the figure and obtain 12. So, $12s$ is the perimeter. Thus, this value is equivalent to 72. So, $s = 6$.

Now, we can retrace our steps. We want to find the area of a single square which is just $s^2 = 6 \cdot 6 = 36$. Then, there are five squares, so we multiply by 5 to obtain $36 \cdot 5 = \boxed{180}$.

※ Missing Pieces

In this section, we dive into a more challenging and thought-provoking type of geometry problem figures with missing pieces. Unlike earlier problems where all measurements were provided and there was some helpful directions to proceed given the formulas, here there is not. These are unique problems that have a special technique which we will walk through. You can't solve them by just plugging in a formula, you have to think strategically and piece together clues like a puzzle.

A rectangle 4 centimeters by 12 centimeters is divided into four triangles, as shown. The areas of three of the four triangles are stated in the diagram. Find the number of square centimeters in the area of the shaded triangle.



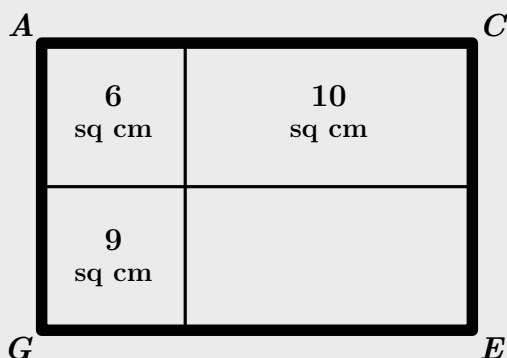
Notice, we want to find the area of the shaded region. But, we are able to find four things: the area of the rectangle, and the area of the three triangles. But, all four triangles collectively form the area of the rectangle. So, it just suffices to find the area of the rectangle, then subtract the area of the three triangles!

Recall, the area of a rectangle is just the product of its two adjacent sides, so the rectangle in this example has an area of $4 \cdot 12 = 48$. Now, we can just subtract the areas of the three triangles which have areas of 8, 16, and 18, so:

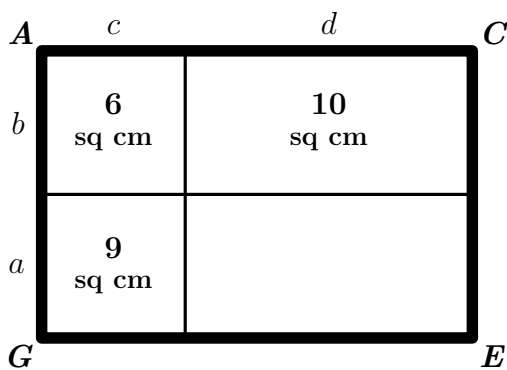
$$48 - 8 - 16 - 18 = \boxed{6}.$$

Before we head into a real challenge, I want to ask you: Do you think we really need to know the dimensions of the rectangle? Short Answer: No! Long Answer: The triangles with areas of 16 and 8 seem to sum to the areas of the shaded region and the triangle with area 18. Do you see why? Try just applying the formula for the area of a triangle!

A rectangle is divided up into four smaller rectangles. The length of each segment is a whole number of centimeters. The areas of three of the rectangles are given in the diagram. Find the number of square centimeters in the total area of rectangle $ACEG$.



Once again, this problem isn't natural in the way that there isn't an obvious solution path. There isn't much given information and even with the areas of the three rectangles it isn't clear how to proceed. However, we do know one thing about rectangles which is the formula for its area. We are also given the area of each of the three rectangles. Lets try and set some variables and write some equations:



Then, using the area of the rectangle formula, we have the following equations:

$$ac = 9, \quad bc = 6, \quad bd = 10.$$

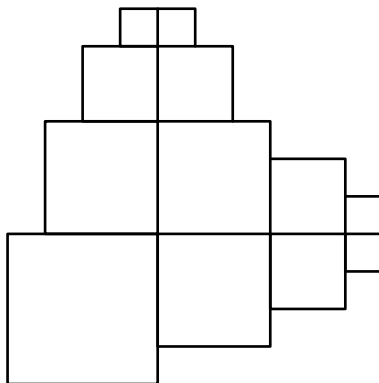
Notice, we want to find ad . Here, we must apply a very clever manipulation which is very difficult to see:

$$ad = \frac{(ac)(bd)}{bc} = \frac{9 \cdot 10}{6} = \boxed{15}.$$

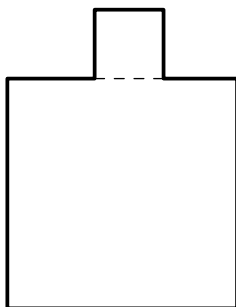
This is a very clever move that doesn't seem to have much motivation. Looking back at the figure, maybe you can see why this is true. This is essentially implying that the product of the opposite rectangles is the same!

※ Practice Problems

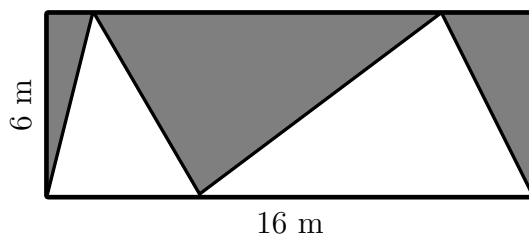
1. The figure shown is built from four 1×1 squares, four 2×2 squares, three 3×3 squares, and one 4×4 square (each measured in centimeters). What is the perimeter of the figure, in cm?



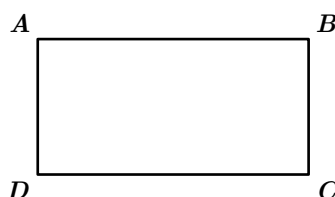
2. A “tower” is formed as shown by placing a small square atop a large square. The perimeter of the tower is 52 centimeters and the perimeter of the large square is 40 centimeters. Find the perimeter of the small square, in centimeters.



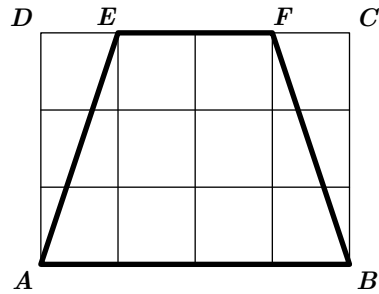
3. What is the area, in square meters, of the shaded part of the rectangle?



4. Lee walks along the edges of a rectangular pool (shown) from point A to B to C to D , a distance of 38 meters. Marina walks along the edges of the same pool from B to C to D to A , a distance of 31 meters. What is the perimeter of the pool, in meters?



5. Let $ABCD$ be a rectangle whose area is 12 square units. How many square units are contained in the area of trapezoid $EFBA$?



6. The figure shown consists of 6 congruent squares. The area of the figure is 96 square centimeters. What is the perimeter of the figure, in centimeters?

